SUPPLEMENTARY MATERIAL

Critical Fragility Emerges from Chromosomal Instability in Cancer

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1 Agent-Based Model Specification: The ODD Protocol

This section provides a standardized, comprehensive description of the **CancerEvo** individual-based simulation framework following the **ODD (Overview, Design concepts, Details) protocol** established by Grimm et al. (2006, 2010). The model simulates the stochastic eco-evolutionary dynamics of cancer cell populations under somatic point mutations, chromosomal instability, and spatial constraints, comparing a spatially structured solid tumor with a well-mixed liquid tumor.

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1.1 Overview

1.1.1 Purpose

The primary purpose of the CancerEvo model is to investigate how the balance between selective growth advantages (oncogenes and tumor suppressor inactivation), mutator phenotypes (point mutation rate and chromosomal missegregation acceleration), and negative purifying selection (lethal homozygous inactivation of essential housekeeping genes) governs the evolutionary trajectories, clonal architecture, and non-equilibrium extinction thresholds of neoplasms. Specifically, the framework addresses:

1. How spatial confinement in solid tissues (local competition within a Moore neighborhood on a 2D regular lattice) versus global fluid mixing in liquid malignancies (mean-field Moran competition) alters the critical mutation threshold separating viable tumor expansion from mutational meltdown (error catastrophe).
2. How chromosomal instability (CIN) and segmental aneuploidy shape karyotypic fragility, buffering capacity, and the transition between diploid, polyploid, and aneuploid evolutionary equilibria.
3. Whether therapeutically accelerating genomic instability can push tumors across the critical extinction boundary into an irreversible collapse phase.

1.1.2 Entities, State Variables, and Scales

Individual Cells: The primary active agents are individual biological cells occupying discrete sites within the tissue. Each cell i is characterized by the following categorical and phenotypic state variables:

- **Cellular State**: $\text{state}_i \in \{0, 1, 2\}$, where:
  - $\text{state}_i = 0$: Healthy wild-type (WT) cell, dividing at homeostatic baseline rate r_0 with zero intrinsic mutation ($\mu_0 = 0$) and zero missegregation ($m_0 = 0$).
  - $\text{state}_i = 1$: Cancer cell, possessing somatic driver mutations or karyotypic alterations, with evolving phenotypic rates r_i, μ_i, and m_i.
  - $\text{state}_i = 2$: Dead cell / vacant site. Generated when a cell suffers lethal mutational load or karyotypic nullisomy/overload ($r_i = 0, \mu_i = 0, m_i = 0$). A dead site exerts zero competitive resistance and is immediately available for colonization by dividing neighbors.
- **Proliferation Rate**: $r_i \in [r_0, r_{\max}]$, representing the probability per unit time of attempting cell division. Wild-type cells divide at $r_0 = 0.15$, while cancer cells increase their proliferation rate in steps of $\Delta r = 0.008$ per activated driver, capped at $r_{\max} = 0.30$.
- **Point Mutation Rate**: $\mu_i \in [0, 1]$, the per-division probability of acquiring a point mutation at any present, unmutated gene locus. Cancer cells advance their mutation rate in increments of $\Delta\mu$ per inactivated mutator gene.

- **Missegregation Rate**: $m_i \in [0, 1]$, the probability per division of undergoing asymmetric chromosomal non-disjunction or segmental translocation, advancing in increments of Δm per inactivated missegregation gene.

- **Ploidy (Chromosome Copy Number)**: $N_{\mathrm{CHR},i} \in \{1, 2, \ldots, 5\}$, denoting the total number of physical chromosome homologues present in cell i. Cells are initialized as diploid ($N_{\mathrm{CHR}} = 2$), except in designated haploid stability sweeps ($N_{\mathrm{CHR}} = 1$). A cell experiencing nullisomy ($N_{\mathrm{CHR}} = 0$) or karyotypic overload ($N_{\mathrm{CHR}} > 5$) dies immediately.

Genomic Architecture and Locus Representation: Each chromosome copy $c \in \{1, \ldots, N_{\mathrm{CHR}}\}$ within cell i contains $N_{\mathrm{genes}} = 45$ functional gene loci, represented computationally by high-performance 64-bit integer bitmasks:

- $\mathbf{C}_{c,i} \in \{0, 1\}^{N_{\mathrm{genes}}}$ (`chromosomes`): Bitmask tracking somatic point mutations. A bit value of 0 indicates an intact wild-type allele; a bit value of 1 indicates a mutated allele.

- $\mathbf{P}_{c,i} \in \{0, 1\}^{N_{\mathrm{genes}}}$ (`presence`): Bitmask tracking physical gene locus presence. A bit value of 1 indicates that the locus is physically present on chromosome c; a bit value of 0 indicates that the locus has been deleted (via segmental or whole-chromosome loss).

The 45 loci are partitioned into five functional classes:

1. **Instability / Mutator Genes** ($N_I = 10$, bits 0–9): Recessive loci whose inactivation impairs DNA repair and increases the point mutation rate μ.

2. **Oncogenes** ($N_O = 10$, bits 10–19): Dominant driver loci whose mutational activation increases cellular proliferation rate r.

3. **Tumor Suppressor Genes** ($N_S = 10$, bits 20–29): Recessive driver loci whose complete bi-allelic loss/inactivation increases cellular proliferation rate r.

4. **Missegregation Genes** ($N_M = 5$, bits 30–34): Recessive loci whose inactivation impairs mitotic spindle checkpoints and increases chromosomal missegregation rate m.

5. **Essential Housekeeping Genes** ($N_{HK} = 10$, bits 35–44): Strictly essential loci. Complete bi-allelic inactivation or deletion of any single housekeeping gene triggers deterministic cellular death.

Spatial and Temporal Scales:

- **Solid Tumor Environment**: A two-dimensional regular square lattice of size $L \times L$ sites with rigid non-periodic boundaries ($N = L^2 = 6,400$ for bifurcation/phase diagrams, $N = 40,000$ for ensemble and intervention studies). Spatial interactions

and competitive displacement occur within the local 8-cell Moore neighborhood (radius $R = 1$). A single lattice site represents the spatial footprint of one biological cell ($\sim$ 10–20 μm in diameter), corresponding to a modeled tissue patch of 0.8×0.8 mm to 2.0×2.0 mm.

- **Liquid Tumor Environment**: A non-spatial, well-mixed population of fixed carrying capacity $N = L^2$. Any dividing cell interacts globally with cells sampled uniformly at random across the entire tissue volume, representing circulating leukemias or ascites.

- **Temporal Scale**: Time progresses in discrete simulation steps $t = 1, 2, \ldots, n_{\text{steps}}$ (up to $100,000$ steps for long-term critical slowing down investigations). Each step represents an epoch of stochastic division and replacement attempts scaled to cellular generation times.

1.1.3 Process Overview and Scheduling

Within each discrete simulation time step, the following processes occur sequentially:

1. **Sampling of Reproducing Cells**: The total tissue proliferation potential $R = \sum_{i=1}^{N} r_i$ is evaluated. The number of division events K occurring in the timestep is drawn from a Poisson distribution:

$$K \sim \text{Poisson}(R), \quad \text{with } K \leq \sum_{i=1}^{N} \mathbb{I}(r_i > 0) \tag{1}$$

   A set of K distinct mother cells is sampled without replacement with probabilities proportional to their division rates $w_i = r_i/R$.

2. **Spatial Neighbor Selection and Competitive Replacement**: Each sampled mother cell i attempts reproduction:

   - *Solid Model*: Cell i inspects its Moore neighborhood $\mathcal{N}_i$. If cell i is completely surrounded by healthy wild-type cells ($\text{state}_k = 0$ for all $k \in \mathcal{N}_i$), the division event is suppressed (simulating homeostatic contact inhibition). Otherwise, replacement is resolved in two hierarchical steps:
     (a) **Dead Site Colonization**: If dead cell sites (state = 2, $r = 0$) exist in $\mathcal{N}_i$, cell i targets a vacant site with probability $n_{\text{dead}}/n_{\text{neigh}}$ and colonizes it deterministically ($P_{\text{replace}} = 1.0$).
     (b) **Moran Contact Competition**: If no dead site is targeted, a living neighbor $j \in \mathcal{N}_i$ is selected uniformly at random. Cell i displaces neighbor j with symmetric Moran probability:

$$P_{\text{replace}}(i \to j) = \frac{r_i}{r_i + r_j} \tag{2}$$

     If the replacement check fails, the division attempt is aborted.

- *Liquid Model*: A candidate target site j is drawn uniformly from the entire population ($j \neq i$). If site j is dead, replacement succeeds unconditionally. If site j is living, replacement succeeds with Moran probability $r_i/(r_i + r_j)$.

3. **Clonal Replication and Daughter Placement**: Upon successful replacement, the mother cell remains at site i, while the newly generated daughter cell occupies target site j. The daughter initially inherits the exact karyotype, presence mask, and rates of the mother.

4. **Point Mutation**: Both mother cell i and daughter cell j undergo independent somatic mutation. For every chromosome copy $c \in \{1, \ldots, N_{\text{CHR}}\}$ and each locus $g \in \{0, \ldots, N_{\text{genes}} - 1\}$, if the locus is physically present ($\mathbf{P}_c[g] = 1$) and currently unmutated ($\mathbf{C}_c[g] = 0$), it mutates ($\mathbf{C}_c[g] \leftarrow 1$) with independent Bernoulli probability μ.

5. **Chromosomal Missegregation**: With probability m_i (the mother's missegregation rate), an asymmetric segregation error occurs between mother i and daughter j:

   - *Whole-Chromosome Mode (Polyploid / Numerical Aneuploidy)*: A chromosome copy is randomly selected from the mother and transferred to the daughter.
   - *Chunk Mode (Segmental Aneuploidy)*: A contiguous slice of loci of random length $\ell \in [3, N_{\text{genes}} - 3]$ is deleted from one of the mother's homologues and appended to the daughter's lowest available chromosome slot.

   The mother's chromosome array is compacted, and daughter chromosome slots are packed up to the maximum capacity of 6.

6. **Phenotypic Rate Update**: The mutation rate μ, division rate r, and missegregation rate m of both mother and daughter are updated from their genomic bitmasks following the genotype-to-phenotype mapping rules (see Submodels).

7. **Viability Check and Cell Death**: Both cells are evaluated for lethal genetic load. A cell suffers deterministic cell death if:

   - Any essential housekeeping gene (HK) has lost all functional wild-type alleles (all copies mutated or deleted).
   - The cell has suffered complete nullisomy ($N_{\text{CHR}} = 0$) or excessive karyotypic burden ($N_{\text{CHR}} > 5$).

   Dying cells are converted to dead state (state $= 2, r = 0, \mu = 0, m = 0$), clearing their genomic bitmasks to create a vacant lattice site.

1.2 Design Concepts

Basic Principles: The model integrates classical theoretical paradigms:

- **Knudson's Two-Hit Hypothesis**: Tumor suppressor genes, mutator genes, missegregation genes, and housekeeping genes require bi-allelic (or multi-allelic under polyploidy) inactivation to manifest a phenotypic alteration.
- **Mutator Phenotype Hypothesis (Loeb)**: Somatic mutator mutations accelerate evolutionary search through driver fitness landscapes at the cost of escalating lethal mutational burden.
- **Error Catastrophe / Mutational Meltdown (Eigen)**: When the somatic mutation rate exceeds a critical threshold, negative selection acting on essential genes destroys clonal viability, driving the population into extinction.
- **Spatial Eco-Evolutionary Competition**: Contact inhibition and local Moran dynamics in solid tissues generate mechanical boundary effects absent in well-mixed regimes.

Emergence: Complex macro-scale behaviors emerge autonomously from local micro-scale interactions:

- Spatial radial expansion, boundary-localized growth, and necrotic or quiescent cores.
- Clonal interference and hitchhiking of passenger mutations during selective sweeps.
- Non-equilibrium second-order phase transitions in solid tumors versus bistable first-order transitions in liquid tumors.
- Critical slowing down, characterized by variance divergence and autocorrelation peaking near the error catastrophe boundary.
- Emergent karyotype stabilization where aneuploid clones equilibrate around viable ploidies ($N_{\mathrm{CHR}} = 2$–3) avoiding lethal housekeeping nullisomy.

Adaptation and Objective: Individual cells do not exhibit purposeful or cognitive adaptation. Cellular adaptation is strictly Darwinian: cells possessing mutations that confer elevated proliferation rates r_i have an increased likelihood of being sampled for division and winning competitive replacement against neighbors, subject to surviving the housekeeping viability filter.

Sensing and Interaction: Dividing cells sense the physical presence (living vs. dead) and replication rates of their immediate Moore neighbors (in solid tumors) or globally sampled partners (in liquid tumors). Physical interactions are mediated entirely through space competition (displacing neighbors) and opportunistic colonization of vacant dead-cell sites.

Stochasticity: Stochasticity is explicitly modeled at all microscopic stages:

- Poisson generation of total division counts per timestep.
- Fitness-weighted sampling of dividing cells without replacement.
- Uniform random selection of target neighbors within the Moore neighborhood.
- Moran acceptance probability for replacement events.
- Bernoulli trials for point mutations across present, unmutated loci.
- Random slice coordinates (start bit and length) during chunk missegregation.
- Stochastic spatial localization of initial tumor seed cells.

Collectives and Observation: Cells form clonal collectives sharing identical mutational profiles or karyotypes. At the population level, the model tracks global tumor density $f_C(t)$, dead cell density $f_{\text{dead}}(t)$, mean mutation rate $\langle \mu(t) \rangle$, mean division rate $\langle r(t) \rangle$, mean missegregation rate $\langle m(t) \rangle$, effective ploidy distributions $P(N_{\text{CHR}})$, locus-specific activation frequencies, and extinction times T_{ext}.

1.3 Details

1.3.1 Initialization

Simulations are initialized on a tissue grid of $L \times L$ sites (or global pool of size $N = L^2$):

- **Background Healthy Tissue**: All N sites are initialized as healthy wild-type cells (state $= 0, N_{\text{CHR}} = 2$). All 45 loci on both chromosome homologues are physically present ($\mathbf{P} = \text{mask_all}$) and unmutated ($\mathbf{C} = 0$), yielding baseline division rate $r_{\text{WT}} = r_0 = 0.15$, mutation rate $\mu_0 = 0.0$, and missegregation rate $m_0 = 0.0$.
- **Cancer Seed Placement**:
  - *Solid Model*: Cancer seed cells are placed as a compact circular cluster centered at $(L/2, L/2)$ with radius $R_{\text{pert}} = r_{\text{pert}} \cdot L$. Default radius is $r_{\text{pert}} = 0.005$ (~ 5 cells) for ensemble runs, $r_{\text{pert}} = 0.05$ (~ 314 cells) for phase diagrams and interventions, and 10 cells for bifurcation runs (see Table 4).
  - *Liquid Model*: Cancer seed cells ($n_{\text{seed}} = 10$ or 50) are distributed uniformly at random across the tissue.
- **Seed Genotype**: Seed cells are assigned homozygous mutations on both homologues at locus I_1 (bit 0) and locus O_1 (bit 10), with all other 43 loci unmutated. This immediately activates one instability gene ($k_I = 1 \implies \mu_{\text{seed}} = \mu_0 + \Delta\mu$) and one oncogene ($k_O = 1 \implies r_{\text{seed}} = r_0 + \Delta r$), establishing the founding malignant clone.

1.3.2 Input Data

The model is an autonomous, self-contained system and does not rely on exogenous environmental time-series data. In therapeutic intervention experiments, an external perturbation protocol is triggered dynamically when tumor density reaches $f_C(t) \geq 0.30$ (detailed in Section 3, Table 5).

1.3.3 Submodels

Submodel 1: Genotype-to-Phenotype Mapping (Boolean Locus Logic) For cell i with N_{CHR} chromosomes, let $P_{c,g} \in \{0,1\}$ and $C_{c,g} \in \{0,1\}$ denote the presence and mutation status of locus $g \in \{0,\ldots,N_{\text{genes}}-1\}$ on chromosome $c \in \{1,\ldots,N_{\text{CHR}}\}$. A locus possesses a functional wild-type allele if at least one chromosome copy carries a present, unmutated version:

$$\text{FuncWT}(g) = \bigvee_{c=1}^{N_{\text{CHR}}} (P_{c,g} \wedge \neg C_{c,g}) \tag{3}$$

Consequently, a recessive locus is completely inactivated if no chromosome carries a functional wild-type allele:

$$\text{Inactivated}(g) = \neg\text{FuncWT}(g) = \bigwedge_{c=1}^{N_{\text{CHR}}} (\neg P_{c,g} \vee C_{c,g}) \tag{4}$$

Active gene counts are computed according to biological dominance rules:

- **Dominant Oncogenes** ($\mathcal{O}$, bits 10–19): An oncogene is activated if *any* present copy is mutated:

$$k_O = \sum_{g\in\mathcal{O}} \bigvee_{c=1}^{N_{\text{CHR}}} (P_{c,g} \wedge C_{c,g}) \tag{5}$$

- **Recessive Tumor Suppressors** ($\mathcal{S}$, bits 20–29):

$$k_S = \sum_{g\in\mathcal{S}} \text{Inactivated}(g) \tag{6}$$

- **Recessive Instability / Mutator Genes** ($\mathcal{I}$, bits 0–9):

$$k_I = \sum_{g\in\mathcal{I}} \text{Inactivated}(g) \tag{7}$$

- **Recessive Missegregation Genes** ($\mathcal{M}$, bits 30–34):

$$k_M = \sum_{g\in\mathcal{M}} \text{Inactivated}(g) \tag{8}$$

Phenotypic rates are evaluated as:

$$r_i = \min\left(r_{\max},\ r_0 + (k_O + k_S)\cdot\Delta r\right), \quad \mu_i = \mu_0 + k_I \cdot \Delta\mu, \quad m_i = k_M \cdot \Delta m \tag{9}$$

where baseline parameters $(r_0, \Delta r, r_{\max}, \mu_0)$ are listed in Table 3.

Submodel 2: Lethal Mutational Load and Cell Death Cell death is evaluated deterministically. A cell is non-viable and converted to dead state ($\text{state}_i \leftarrow 2$) if it satisfies any of the following biological lethality criteria:

$$[\exists g \in \mathcal{HK} \quad \text{such that} \quad \text{Inactivated}(g) = 1] \quad \vee \quad N_{\text{CHR},i} = 0 \quad \vee \quad N_{\text{CHR},i} > 5 \quad (10)$$

Application to Aneuploid and Heteroploid Cells: The presence-aware Boolean logic ($\text{Inactivated}(g) = \bigwedge_c [\neg P_{c,g} \vee C_{c,g}]$) governs cell fate across numerical and segmental aneuploidy in three distinct operational regimes:

1. **Locus Nullisomy (Segmental or Whole-Chromosome Loss)**: If a cell loses an allele of housekeeping gene g through chunk deletion or chromosome loss, the locus becomes hemizygous ($P_{c,g} = 1$ on only one homologue). If that remaining copy is subsequent deleted ($\forall c, P_{c,g} = 0$), the locus suffers physical nullisomy, immediately rendering $\text{Inactivated}(g) = 1$ and triggering instantaneous cell death.

2. **Hemizygous Vulnerability**: In a cell that is hemizygous for housekeeping gene g (due to prior segmental deletion or monosomy, $N_{\text{CHR}} = 1$), a single somatic point mutation on the lone surviving allele ($C_{1,g} \leftarrow 1$) eliminates the only functional copy ($\text{FuncWT}(g) = 0$), causing cell death. In contrast to diploid cells (which require two hits), hemizygous clones possess zero buffering capacity.

3. **Polyploid Buffering and Cryptic Damage**: In polysomic or polyploid cells ($N_{\text{CHR}} \in \{3, 4, 5\}$), a housekeeping locus remains functional as long as at least one intact copy survives. These cells can accumulate deleterious mutations on multiple copies without dying (buffering capacity). However, if subsequent missegregation transfers away or deletes the single remaining functional copy, the cell is left exclusively with mutated alleles, triggering delayed catastrophic collapse (karyotypic fragility).

4. **Global Ploidy Bounds**: A cell dies immediately if it suffers total chromosome loss ($N_{\text{CHR}} = 0$, complete nullisomy) or exceeds the maximum viable karyotype limit ($N_{\text{CHR}} > 5$, representing proteotoxic stress and severe mitotic catastrophe).

Crucially, lethality is evaluated twice within every division event: first immediately following point mutation, and re-evaluated immediately following chromosome/chunk transfer for both mother and daughter cells. If a transfer deprives the mother of its last functional housekeeping allele or pushes the daughter's ploidy beyond 5, that cell is converted to dead site ($\text{state} = 2, r = 0, \mu = 0, m = 0$), clearing all genomic arrays.

Submodel 3: Moran Competition and Replacement Dynamics When dividing mother cell i targets neighbor site j:

$$P_{\text{replace}}(i \to j) = \begin{cases} 1.0, & \text{if site } j \text{ is dead } (\text{state}_j = 2, r_j = 0) \\ \dfrac{r_i}{r_i + r_j}, & \text{if site } j \text{ is living } (\text{state}_j \in \{0, 1\}) \end{cases} \quad (11)$$

In the solid model, if dead neighbors exist in the Moore neighborhood $\mathcal{N}_i$, dead sites are targeted with probability $n_{\text{dead}}/n_{\text{neigh}}$ to prioritize filling tissue voids before displacing living neighbors.

Submodel 4: Chromosomal Missegregation Mechanics When missegregation occurs (with probability m_i):

- **Whole-Chromosome Transfer**: A random chromosome $c_{\text{donor}} \in \{1, \ldots, N_{\text{CHR},i}\}$ is selected from mother i. The full chromosome (mutations and presence masks) is removed from mother i and transferred into the lowest unoccupied slot of daughter j.
- **Chunk-Based Segmental Missegregation**: A cut length $\ell \in \{3, \ldots, N_{\text{genes}} - 3\}$ and start locus $g_{\text{start}} \in \{0, \ldots, N_{\text{genes}} - 1\}$ are chosen uniformly at random. A contiguous bitwise mask of length ℓ with periodic wrap-around is defined:

$$\mathbf{M}_{\text{chunk}} = \bigvee_{b=0}^{\ell-1} (1 \ll [(g_{\text{start}} + b) \pmod{N_{\text{genes}}}]) \tag{12}$$

  The slice $\mathbf{S} = \mathbf{M}_{\text{chunk}} \wedge \mathbf{P}_{c_{\text{donor}},i}$ is removed from the mother's homologue ($\mathbf{P}_{c_{\text{donor}},i} \leftarrow \mathbf{P}_{c_{\text{donor}},i} \wedge \neg\mathbf{S}$, $\mathbf{C}_{c_{\text{donor}},i} \leftarrow \mathbf{C}_{c_{\text{donor}},i} \wedge \neg\mathbf{S}$) and integrated into the daughter cell across available slots $c \in \{1, \ldots, 6\}$.

Following transfer, the mother cell compacts its remaining chromosome slots, and both mother and daughter re-evaluate their viability and phenotypic rates.

2 Analytical Master Equation and Stationary Mutational Distributions in Liquid Tumors

We consider a liquid tumor model with fixed population size N. The number of wild-type cells is indicated with W, the number of dead cells with D. The cancer cell population can be further divided in $N_{\mathcal{I}}$ sub-populations (one per instability class), with $N_{\mathcal{I}}$ the number of instability genes. We indicate the abundance of each cancer sub-population with C_k, where $k \in \{1, \ldots, N_I\}$ counts the number of mutated instability genes. We work with densities (fractions) rather than absolute abundances:

$$f_W \equiv \frac{W}{N}, \quad f_k \equiv \frac{C_k}{N}, \quad p_D \equiv \frac{D}{N} \tag{13}$$

These are constrained by the fixed-size normalization:

$$\sum_{k=1}^{N_I} f_k + f_W + p_D = 1 \tag{14}$$

Survival probability after a mutation round

Both cancer and wild-type cells undergo replication followed by a stochastic mutation event. After mutation, a viability check is performed on all N_{HK} housekeeping genes: if any housekeeping gene becomes fully mutated (on all chromosome copies), the cell dies. Each gene has an independent per-locus mutation probability $\mu_k = k \cdot \delta\mu$ for a class-k cell. The probability that a single housekeeping gene survives (i.e. is not mutated) in one round is $(1 - k\,\delta\mu)$. Since all N_{HK} housekeeping genes must independently survive:

$$P_{s,k} = \underbrace{(1 - k\,\delta\mu)}_{\text{one HK gene survives}} \times \cdots \times \underbrace{(1 - k\,\delta\mu)}_{N_{HK}\text{ times}} = (1 - k\,\delta\mu)^{N_{HK}} \tag{15}$$

The post-replication fates (exact multi-step model)

After a mutation round, a cell in class k can either remain in class k (cloning), promote to class $k + j$ with $j \geq 1$ (multi-step promotion), or die due to a housekeeping mutation. Since mutations at different loci occur independently with per-locus probability $\mu_k = k \cdot \delta\mu$, we derive these probabilities exactly:

1. Faithful cloning (α_k): The cell remains in class k. This requires that none of the N_{HK} housekeeping genes and none of the $(N_I - k)$ remaining unmutated instability genes are hit. The total number of loci that must remain unmutated is $N_{HK} + N_I - k$:

$$\alpha_k = (1 - k\,\delta\mu)^{N_{HK}+N_I-k} \tag{16}$$

2. Promotion ($P_{k\to k+j}$): The cell advances from class k to class $k+j$ (where $j \in \{1,\ldots,N_I-k\}$ instability genes mutate, while all other instability loci and all housekeeping loci remain unmutated). Since the selection of which instability genes mutate is combinatorial:

$$P_{k\to k+j} = \binom{N_I-k}{j}(k\,\delta\mu)^j(1-k\,\delta\mu)^{N_{HK}+N_I-k-j} \tag{17}$$

Note that faithful cloning is simply the diagonal case $\alpha_k = P_{k\to k}$.

3. Lethal mutation (δ_k): The cell dies because at least one housekeeping gene is hit, which is the complement of the survival probability $P_{s,k} = (1-k\,\delta\mu)^{N_{HK}}$:

$$\delta_k = 1 - P_{s,k} = 1 - (1-k\,\delta\mu)^{N_{HK}} \tag{18}$$

The sum of all possible survival outcomes plus the death probability is exactly 1:

$$\sum_{j=0}^{N_I-k} P_{k\to k+j}+\delta_k = (1-k\,\delta\mu)^{N_{HK}} \sum_{j=0}^{N_I-k}\binom{N_I-k}{j}(k\,\delta\mu)^j(1-k\,\delta\mu)^{N_I-k-j}+[1-(1-k\,\delta\mu)^{N_{HK}}] = 1 \tag{19}$$

This binomial formulation provides an exact description of the mutation process without needing leading-order approximations.

Moran competition probabilities

When a dividing cell targets a living (non-dead) cell, the outcome is decided by a Moran process: the divider replaces the target with probability proportional to its fitness relative to the combined fitness. All cancer classes share the same intrinsic division rate r_m, while wild-type cells divide at rate r_0:

- **Cancer vs Wild-Type:** The cancer cell (rate r_m) competes against a wild-type cell (rate r_0):
$$\beta_{cw} = \frac{r_m}{r_m + r_0} \tag{20}$$
- **Cancer vs Cancer:** Two cancer cells with the same rate r_m compete symmetrically:
$$\beta_{cc} = \frac{r_m}{r_m + r_m} = \frac{1}{2} \tag{21}$$
- **Wild-Type vs Cancer:** The complement of the cancer-vs-WT outcome:
$$\beta_{wc} = \frac{r_0}{r_m + r_0} = 1 - \beta_{cw} \tag{22}$$

System parameters are formalized in Table 1.

Table 1: Systematic Overview of the Unified Parameter Space

Symbol	Biological Definition	Functional Form / Constraint
N	Total fixed tissue lattice sites	Constant volume constraint
N_I	Maximum viable cancer mutational classes	Total mutable Instability genes
N_{HK}	Total critical Housekeeping genes	Continuous fitness check loci
$\delta\mu$	Per-locus mutation rate increment	Constant scaling factor
r_m	Intrinsic division rate of class k cancer	Phenotypic fitness parameter
r_0	Active division rate of healthy wild-type	Homeostatic driver baseline
α_k	Probability of a faithful clone	$(1-k\,\delta\mu)^{N_{HK}+N_I-k}$
$P_{k\to k+j}$	Probability of promotion ($k \to k+j$)	$\binom{N_I-k}{j}(k\,\delta\mu)^j(1-k\,\delta\mu)^{N_{HK}+N_I-k-j}$
δ_k	Lethal mutation probability	$1-(1-k\,\delta\mu)^{N_{HK}}$
β_{cw}	Moran win probability of cancer vs WT	$r_m/(r_m+r_0)$
β_{cc}	Moran win probability of cancer k vs j	$r_m/(r_m+r_m)$
β_{wc}	Moran win probability of WT vs cancer k	$r_0/(r_m+r_0) = 1-\beta_{cw}$

2.1 Microscopic Mechanisms and Elementary Transition Rates

After each duplication step, both mother and daughter cells independently undergo stochastic mutation events as described above. Both cells independently draw from the same probability distribution: cloning (α_k), promoting ($P_{k\to k+j}$), or dying (δ_k). Crucially, DNA replication and subsequent viability/mutation checks are only performed upon successful division (Moran competition success). If division is aborted, neither the mother nor the daughter replicates or mutates. Thus, all division-dependent rates (cloning, promotion, and death) for both mother and daughter cells are scaled by the division success probability ϕ_k.

Effective reachable space ϕ_k

We define the fractional reachable space ϕ_k as the effective probability that a dividing class-k cancer cell successfully places its daughter. The daughter can land on three types of targets:

$$\phi_k = \underbrace{p_D}_{\substack{\text{dead slots:}\\ \text{always available}}} + \underbrace{f_W\,\beta_{cw}}_{\substack{\text{WT cells displaced}\\ \text{via Moran competition}}} + \underbrace{\sum_{j=1}^{N_I} f_j\,\beta_{cc}}_{\substack{\text{cancer cells displaced}\\ \text{(50\% win rate)}}} \tag{23}$$

- p_D: the fraction of dead (empty) slots. Replacement always succeeds (no competition needed).
- $f_W\beta_{cw}$: wild-type cells are encountered with frequency f_W, and the cancer cell wins the Moran competition with probability β_{cw}.

- $\sum_j f_j \beta_{cc}$: other cancer cells are encountered with total frequency $\sum_j f_j$, with a symmetric 50% win rate $\beta_{cc} = 1/2$.

Elementary transitions

We now enumerate every possible microscopic event and its rate. The key insight is that during a single division event of a class-k cancer cell, two independent fates are resolved: (i) the **mother fate** (what happens to the original cell on its site after mutation), and (ii) the **daughter fate** (what happens to the copy placed at the target site after mutation). Both fates are conditioned on the division being successfully completed.

The complete transition rate matrix is given in Table 2. We explain each row below.

Table 2: Complete Microscopic Transition Rate Matrix (Symmetrical Hazards)

Active Vector	**Target & Event Type**	$\mathbf{\Delta C_k}$	$\mathbf{\Delta D}$	**Transition Rate** (λ)
Cancer Class k	**Mother Fate:** Dies on Site	-1	$+1$	$C_k\, r_m\, \phi_k\, \delta_k$
Cancer Class k	**Mother Fate:** Promotes on Site ($k \to k+j$)	$-1_k, +1_{k+j}$	0	$C_k\, r_m\, \phi_k\, P_{k\to k+j}$
Cancer Class k	**Daughter Fate:** Clones on D	$+1$	-1	$C_k\, r_m\, p_D\, \alpha_k$
Cancer Class k	**Daughter Fate:** Promotes on D ($k \to k+j$)	$+1_{k+j}$	-1	$C_k\, r_m\, p_D\, P_{k\to k+j}$
Cancer Class k	**Daughter Fate:** Dies on D	0	0	$C_k\, r_m\, p_D\, \delta_k$
Cancer Class k	**Daughter Fate:** Clones on W	$+1$	0	$C_k\, r_m\, f_W\, \beta_{cw}\, \alpha_k$
Cancer Class k	**Daughter Fate:** Dies on W	0	$+1$	$C_k\, r_m\, f_W\, \beta_{cw}\, \delta_k$
Wild-Type (W)	WT Divides, Replaces Dead Void	0	-1	$W\, r_0\, p_D$
Wild-Type (W)	WT Divides, Clears Cancer k	-1	0	$W\, r_0\, f_k\, \beta_{wc}$

2.2 Mean-Field Ordinary Differential Equations

By taking the thermodynamic limit ($N \to \infty$), the stochastic master equation converges to a deterministic system of ODEs. We derive the ODE for each compartment by summing all transition rates from Table 2 that affect that compartment, converting from absolute counts to densities ($f_k = C_k/N$).

Interior cancer classes ($1 < k < N_I$)

For an interior cancer class k ($1 < k < N_I$), the rate of change of the cell fraction f_k is determined by five primary biological processes: (i) net clonal growth and survival ($r_m f_k \phi_k (2\alpha_k - 1)$), which balances faithful daughter cloning against mother cell loss via death or promotion; (ii) self-displacement loss within the same class ($r_m f_k^2/2$); (iii) cross-class loss due to displacement by other cancer classes ($f_k \sum_{j\neq k} r_m f_j \beta_{cc}$); (iv) wild-type clearance loss ($r_0 f_W f_k \beta_{wc}$); and (v) promotion influx from all upstream classes $i < k$ ($\sum_{i=1}^{k-1} 2\, r_m f_i \phi_i P_{i\to k}$). Combining these contributions yields:

$$\frac{df_k}{dt} = r_m f_k \phi_k (2\alpha_k - 1) - r_m \frac{f_k^2}{2} - f_k \sum_{j\neq k} r_m f_j \beta_{cc} - r_0 f_W f_k \beta_{wc} + \sum_{i=1}^{k-1} 2\, r_m f_i \phi_i P_{i\to k} \quad (24)$$

Boundary class $k = 1$

The founding cancer class $k = 1$ receives no upstream mutational influx (there is no class $k = 0$). The equation is identical to the interior case but without Term 6:

$$\frac{df_1}{dt} = r_m f_1 \phi_1 (2\alpha_1 - 1) - r_m \frac{f_1^2}{2} - f_1 \sum_{j \neq 1} r_m f_j \beta_{cc} - r_0 f_W f_1 \beta_{wc} \tag{25}$$

Terminal class $k = N_I$

For the terminal mutational class, all instability genes are already mutated, so further promotion is impossible: $P_{N_I \to N_I + j} = 0$. The mother-loss term simplifies to $-r_m f_{N_I} \phi_{N_I} (1 - \alpha_{N_I})$ where $1 - \alpha_{N_I} = \delta_{N_I}$. The promotion influx from all classes $i < N_I$ is present:

$$\frac{df_{N_I}}{dt} = r_m f_{N_I} \phi_{N_I} (2\alpha_{N_I} - 1) - r_m \frac{f_{N_I}^2}{2} - f_{N_I} \sum_{j \neq N_I} r_m f_j \beta_{cc} - r_0 f_W f_{N_I} \beta_{wc} + \sum_{i=1}^{N_I - 1} 2\, r_m f_i \phi_i P_{i \to N_I} \tag{26}$$

Wild-type compartment f_W

The WT population grows when WT cells divide and successfully place daughters on dead slots or displace cancer cells, and shrinks when cancer daughters displace WT cells:

- **Gain:** WT divides onto dead slots (rate $r_0 f_W p_D$) or displaces cancer class j (rate $r_0 f_W f_j \beta_{wc}$ per class).
- **Loss:** Cancer class k daughters displace WT cells (rate $r_m f_k f_W \beta_{cw}$ per class).

$$\frac{df_W}{dt} = \underbrace{r_0 f_W \left(p_D + \sum_{j=1}^{N_I} f_j \beta_{wc} \right)}_{\text{WT expansion (dead slots + cancer clearance)}} - \underbrace{f_W \sum_{k=1}^{N_I} r_m f_k \beta_{cw}}_{\text{displacement by cancer daughters}} \tag{27}$$

Dead compartment p_D

Dead cells accumulate from lethal mutations (both mother dying in place and daughter dying after placement) and are consumed when any living cell fills a dead slot. Since DNA replication only happens during successful division, both mother and daughter can only die if division succeeds.

- **Gain from mother death:** $r_m f_k \phi_k \delta_k$ per class (row 1).
- **Gain from daughter death:** $r_m f_k \phi_k \delta_k$ per class (combining rows 5, 7, and cancer-cancer targets).

- **Total death-related gain:** $2\, r_m f_k \phi_k \delta_k$ per class.
- **Loss from filling:** Dead slots are consumed by cancer daughters ($r_m f_k p_D$, regardless of fate) and WT daughters ($r_0 f_W p_D$).

$$\frac{dp_D}{dt} = \underbrace{\sum_{k=1}^{N_I} r_m f_k \left[2\phi_k \delta_k - p_D\right]}_{\text{cancer-driven death accumulation minus filling}} - \underbrace{r_0 f_W p_D}_{\text{WT filling dead slots}} \tag{28}$$

2.3 Stationary State Distributions and Stability Analysis

Linear stability of the healthy fixed point

We examine the linear stability of the healthy fixed point ($f_W^* = 1,\ p_D^* = 0,\ f_k^* = 0\ \forall k$) against an infinitesimal perturbation $f_1 = \epsilon \ll 1$, with all other cancer classes still at zero.

Starting from the $k = 1$ boundary ODE and substituting the healthy fixed-point values, each term simplifies as follows:

- **Mother loss & daughter clone (combined):** $r_m\, \epsilon\, \phi_1\, (2\alpha_1 - 1)$. At the fixed point, $\phi_1 \to f_W \beta_{cw} = 1 \cdot \beta_{cw}$. This gives $r_m\, \beta_{cw}\, (2\alpha_1 - 1)\, \epsilon$.
- **Self-displacement:** $-r_m\, \epsilon^2/2 \approx 0$ — vanishes at $\mathcal{O}(\epsilon^2)$.
- **Cross-class loss:** $-\epsilon \sum_{j \neq 1} r_m f_j \beta_{cc} = 0$.
- **WT clearance:** $-r_0\, (1)\, \epsilon\, \beta_{wc} = -r_0\, \beta_{wc}\, \epsilon$.
- **Promotion influx:** absent for $k = 1$.

Collecting all $\mathcal{O}(\epsilon)$ terms:

$$\frac{df_1}{dt} \approx \left[r_m\, \beta_{cw}\, (2\alpha_1 - 1) - r_0\, \beta_{wc}\right] \epsilon \tag{29}$$

Substituting $\beta_{cw} = r_m/(r_m + r_0)$ and $\beta_{wc} = r_0/(r_m + r_0)$:

$$\frac{df_1}{dt} \approx \frac{r_m^2\, (2\alpha_1 - 1) - r_0^2}{r_m + r_0}\, f_1 \tag{30}$$

Derivation of the growth foothold inequality

The tumor can invade when the bracketed growth rate is positive:

$$r_m^2\, (2\alpha_1 - 1) - r_0^2 > 0 \tag{31}$$

This yields the Unified Growth Foothold Inequality:

$$r_m^2\,(2\alpha_1 - 1) > r_0^2 \iff 2\alpha_1 - 1 > \left(\frac{r_0}{r_m}\right)^2 \iff \alpha_1 > \frac{1 + (r_0/r_m)^2}{2} \tag{32}$$

The left-hand side balances the cancer cell's maximum birth rate times its net clonal expansion factor $(2\alpha_1 - 1)$, which accounts for the survival and fidelity of both the mother and daughter during a division event. The right-hand side r_0 represents the WT homeostatic division pressure. When mutation rates are too high, the cancer's net growth rate drops below the wild-type's division rate, preventing the tumor from taking foothold. The quadratic scaling in r_m^2 and r_0^2 emerges because replacement requires both division and competitive Moran displacement, so each cell type's effective expansion power scales with the square of its proliferation rate.

Stationary state

When the growth inequality is satisfied, the system reaches a tumor-dominated stationary state where $df_k/dt = 0$ for all k. Setting the interior ODE to zero and rearranging:

$$0 = r_m f_k^* \phi_k^* (2\alpha_k - 1) - r_m \frac{(f_k^*)^2}{2} - f_k^* \sum_{j \neq k} r_m f_j^* \beta_{cc} - r_0 f_W^* f_k^* \beta_{wc} + \sum_{i=1}^{k-1} 2\, r_m f_i^* \phi_i^* P_{i \to k} \tag{33}$$

We separate the ϕ_k^* sum into its self-interaction part $(f_k^* \beta_{cc})$ and the rest $(\phi_k^{(\neq k)*})$:

$$\phi_k^* = \phi_k^{(\neq k)*} + f_k^*\, \beta_{cc} \tag{34}$$

Substituting and grouping terms by powers of f_k^*:

$$\begin{aligned} 0 &= r_m f_k^* \left[\phi_k^{(\neq k)*} + f_k^* \beta_{cc}\right] (2\alpha_k - 1) - r_m \frac{(f_k^*)^2}{2} \\ &\quad - f_k^* \sum_{j \neq k} r_m f_j^* \beta_{cc} - r_0 f_W^* f_k^* \beta_{wc} + \sum_{i=1}^{k-1} 2\, r_m f_i^* \phi_i^* P_{i \to k} \\ 0 &= r_m f_k^* \phi_k^{(\neq k)*} (2\alpha_k - 1) + r_m (f_k^*)^2 \beta_{cc} (2\alpha_k - 1) - r_m \frac{(f_k^*)^2}{2} \\ &\quad - f_k^* \sum_{j \neq k} r_m f_j^* \beta_{cc} - r_0 f_W^* f_k^* \beta_{wc} + \sum_{i=1}^{k-1} 2\, r_m f_i^* \phi_i^* P_{i \to k} \end{aligned} \tag{35}$$

Since $\beta_{cc} = 1/2$, the quadratic terms combine as:

$$-r_m (1 - \alpha_k)(f_k^*)^2 \tag{36}$$

Collecting terms, this takes the standard quadratic form:

$$A_k\,(f_k^*)^2 - B_k\,f_k^* - Q_{k-1}^* = 0 \tag{37}$$

where the coefficients are:

1. **Quadratic (self-interaction) coefficient (A_k):**

$$A_k = r_m(1-\alpha_k) \tag{38}$$

This arises from the combination of self-displacement ($\beta_{cc} = 1/2$) and the mutation outcomes.

2. **Linear (net growth) coefficient (B_k):**

$$B_k = r_m\,\phi_k^{(\neq k)*}\,(2\alpha_k - 1) - \sum_{j\neq k} r_m f_j^*\,\beta_{cc} - r_0\,f_W^*\,\beta_{wc} \tag{39}$$

This coefficient represents the net per-capita growth rate of class k at the stationary state, excluding self-interaction and upstream promotion.

3. **Upstream promotion source (Q_{k-1}^*):**

$$Q_{k-1}^* = \sum_{i=1}^{k-1} 2\,r_m\,f_i^*\,\phi_i^*\,P_{i\to k} \tag{40}$$

This is the rate at which cells flow into class k from all lower classes $i < k$ (dual-sourced: mother promoting + daughter promoting). For $k = 1$, $Q_0^* = 0$.

Solution via the quadratic formula

The quadratic $A_k x^2 - B_k x - Q_{k-1}^* = 0$ (with $x = f_k^*$) has the physical (positive) root:

$$f_k^* = \frac{B_k + \sqrt{B_k^2 + 4\,A_k\,Q_{k-1}^*}}{2\,A_k} \tag{41}$$

The positive sign before the square root is chosen because $f_k^* \geq 0$ is required. The system is solved sequentially: f_1^* is determined first (with $Q_0^* = 0$), then f_2^* using f_1^*, and so on up to $f_{N_I}^*$.

3 Simulation Parameters and Experimental Configurations

Here are presented the key parameter values and configurations used in the simulation framework, both in the neighborhood-restricted solid tumor model and the global, well-mixed liquid tumor model.

3.1 Genomic Architecture and Baseline Rates

Table 3 outlines the shared genomic architecture and rate parameters that are identical across both the solid and liquid tumor models to ensure direct comparability.

Table 3: Shared Genomic Architecture and Baseline Simulation Rates

Parameter	**Symbol**	**Default Value**	**Biological / Model Description**
Genomic Architecture			
Ploidy Copy Number	N_{CHR}	2[a]	Initial number of chromosome copies per cell
Instability Genes	N_I	10	Loci driving mutation rate increments
Oncogenes	N_O	10	Loci driving division rate increments
Suppressor Genes	N_S	10	Loci driving division rate increments
Missegregation Genes	N_M	5	Loci driving chromosome missegregation
Housekeeping Genes	N_{HK}	10	Essential loci (homozygous mutation is lethal)
Total Loci per Chromosome	N_{genes}	45	Sum of all gene classes ($N_I + N_O + N_S + N_M + N_{HK}$)
Initial Cancer Seed State	–	–	1 mutated I gene (bit 0), 1 mutated O gene (bit 10)[b]
Baseline Rates			
Baseline Division Rate	r_0	0.15	Birth rate of healthy wild-type cells ($r_{\text{WT}} = r_0$)
Driver Rate Increment	dr	0.008	Birth rate increase per mutated O or S allele
Maximum Division Rate	$r_{\max}$	0.30	Upper cap on division rate ($2 \times r_0$)
Baseline Mutation Rate	μ_0	0.0	Background mutation rate per locus per division

[a] Set to $N_{\text{CHR}} = 1$ (haploid) specifically during the stability sweeps.
[b] Represented as chromosome bitmask where all other 43 loci start unmutated.

3.2 Simulation Configurations and Sweep Parameters

Table 4 details the distinct parameter sets and tissue environments used for the four primary studies in both the Solid and Liquid cases: Ensemble Simulations, Stability Sweeps, Parameter Phase Diagrams, and Therapeutic Interventions.

Table 4: Systematic Overview of Simulation Configuration Parameters

Configuration Variable	**Ensemble Simulations**	**Stability Sweeps**	**Parameter Phase Diagrams**	**Therapeutic Interventions**
Solid Tumor Model (Local Moran-like competition in 2D Moore neighborhood)				
Lattice Side (L)	200	200	80	200
Total Cells ($N = L^2$)	40,000	40,000	6,400	40,000
Max Simulation Steps (n_{steps})	2500	500	1000	Up to 3000[a]
Ploidy (N_{CHR})	2	1 (Haploid)	2	2
Seeding Strategy	Circular seed ($R = 0.005L$)	Circular seed ($R \approx 0.252L$)	Circular seed ($R = 0.05L$)	Circular seed ($R = 0.05L$)
Initial Tumor Fraction ($f_C(0)$)	$\sim 0.01\%$ (~ 5 cells)	20.0% ($\sim 8,000$ cells)	$\sim 0.8\%$ (~ 49 cells)	$\sim 0.8\%$ (~ 314 cells)
Mutation Rate Increment ($d\mu$)	0.012	Swept $[10^{-6}, 0.02]$[b]	Swept [0.0001, 0.03]	0.012 (0.024 in D)
Missegregation Rate (dm)	0.0 (D) / 0.01 (A, P)	0.0	0.0	0.0
Missegregation Type	Whole (D, P) / Chunk (A)	Whole	Whole	Whole
Replicate Count	500 per ploidy	1 per $r_{\max}$ step	50 per grid point (20×20)	1 per intervention
Termination Conditions	$f_{\text{WT}} < 0.5$ or $f_C = 0$	$f_{\text{WT}} < 0.76$ or $f_{\text{WT}} > 0.84$	$f_{\text{WT}} < 0.5$ or $f_C = 0$	$f_{\text{WT}} < 0.5$ or $f_C = 0$
Liquid Tumor Model (Global Moran-like competition with random placement)				
Lattice Side (L)	200	200	80	–[c]
Total Cells ($N = L^2$)	40,000	40,000	6,400	–
Max Simulation Steps (n_{steps})	2500	3000	3000	–
Ploidy (N_{CHR})	2	1 (Haploid)	2	–
Seeding Strategy	Scattered random sites	Scattered random sites	Scattered random sites	–
Initial Seed Count (n_{seed})	10 cells	10 cells	50 cells	–
Mutation Rate Increment ($d\mu$)	0.023 (D) / 0.045 (A, P)	Swept $[10^{-6}, 0.04]$[b]	Swept [0.0001, 0.03]	–
Missegregation Rate (dm)	0.0 (D) / 0.01 (A, P)	0.0	0.0	–
Missegregation Type	Whole (D, P) / Chunk (A)	Whole	Whole	–
Replicate Count	500 per ploidy	1 per $r_{\max}$ step	50 per grid point (20×20)	–
Termination Conditions	$f_{\text{WT}} < 0.5$ or $f_C = 0$	$f_C = 1.0$ or $f_C = 0.0$	$f_{\text{WT}} < 0.5$ or $f_C = 0$	–

[a] Pre-intervention phase runs up to 2000 steps until the tumor density $f_C \geq 0.30$ trigger is met; the post-intervention phase runs for exactly 1000 steps.
[b] Refined dynamically using a bisection algorithm with 14 iterations for each division rate step.
[c] Therapeutic interventions are implemented and analyzed exclusively for the solid tumor model.
Note on Ensemble Labels: **D** = Diploid ($N_{\text{CHR}} = 2, dm = 0.0$), **A** = Aneuploid ($N_{\text{CHR}} = 2, dm = 0.01$, chunk-based missegregation), **P** = Polyploid ($N_{\text{CHR}} = 2, dm = 0.01$, whole-chromosome missegregation).

3.3 Therapeutic Intervention Mechanisms (Solid Tumor)

Table 5 describes the targeting rules, execution parameters, and physiological analogues of the four distinct therapeutic interventions simulated in the solid tumor model.

Table 5: Mechanistic Parameters of simulated Therapeutic Interventions

Type	Biological Concept	Model Parameter	Clearance / Mutation Update Rule
A	Fitness-selective targeted therapy	r_i (cell division rate)	Active cancer cells are cleared (reset to WT) with cell-specific probability $P(\text{clear}) = 1 - r_0/r_i$.
B	Driver-selective early clone therapy	$i_{\text{lim}} \leq 1$ ($\mu \leq d\mu$)	Cells belonging to mutation class 0 or 1 are cleared with probability $p_d = 1.0$.
C	Instability-selective late clone therapy	$i_{\text{lim}} \geq 3$ ($\mu \geq 3d\mu$)	Cells belonging to mutation class ≥ 3 are cleared with probability $p_d = 1.0$.
D	Mutagenic load therapy	new $d\mu = 0.024$	The mutation rate increment parameter $d\mu$ is doubled from 0.012 to 0.024, immediately doubling the mutation rates of all living cells.

Common Parameters: Interventions are triggered exactly when the active cancer density $f_C \geq 0.30$ (30% of the tissue area). Displaced or cleared cancer cells are reset to the healthy wild-type state ($r_{\text{cell}} = r_0$, $\mu_{\text{cell}} = \mu_0$, $N_{\text{CHR}} = 2$, chromosomes unmutated).

4 Bifurcation and Critical Transition Analysis

Table 6 details the specific simulation parameters, sweep ranges, tissue capacities, and observed critical thresholds for the bifurcation, extinction time, and critical slowing down analyses presented in Figure 1 of the main manuscript.

Table 6: Simulation Parameters and Critical Values for Bifurcation Analyses (Figure 1)

Panel & Model	Swept Parameter	Sweep Range	Critical Threshold	Dynamical Feature / Regime
(a) Solid $\Delta\mu$	Mutation increment $\Delta\mu$	$[13.0, 21.0] \times 10^{-3}$	$\Delta\mu_c \approx 16.8 \times 10^{-3}$	Continuous transition; critical slowing down peak
(b) Liquid $\Delta\mu$	Mutation increment $\Delta\mu$	$[19.0, 27.0] \times 10^{-3}$	$\Delta\mu_c \approx 22.8 \times 10^{-3}$	Bistability window $[22.7, 23.7] \times 10^{-3}$; slowing down
(c) Solid Δr	Division increment Δr	$[1.0, 7.0] \times 10^{-3}$	$\Delta r_c \approx 3.9 \times 10^{-3}$	Inverse transition (extinction to persistence)
(d) Liquid Δr	Division increment Δr	$[1.0, 4.0] \times 10^{-3}$	$\Delta r_c \approx 2.5 \times 10^{-3}$	Lower persistence threshold under global mixing

Shared Parameters for Bifurcation Runs: Tissue capacity $N = 6,400$ ($L = 80 \times 80$ grid for Solid; global Moran pool for Liquid); initial tumor seed $n_{\text{seed}} = 10$ cancer cells carrying I_1 and O_1 mutations; ploidy $N_{\text{CHR}} = 2$ (diploid); maximum simulation duration $n_{\text{steps}} = 100,000$ steps; extinction threshold $N_{\text{cells}} \leq 40$ ($f_C \leq 0.00625$); replicate count = 10 stochastic realizations per parameter step.

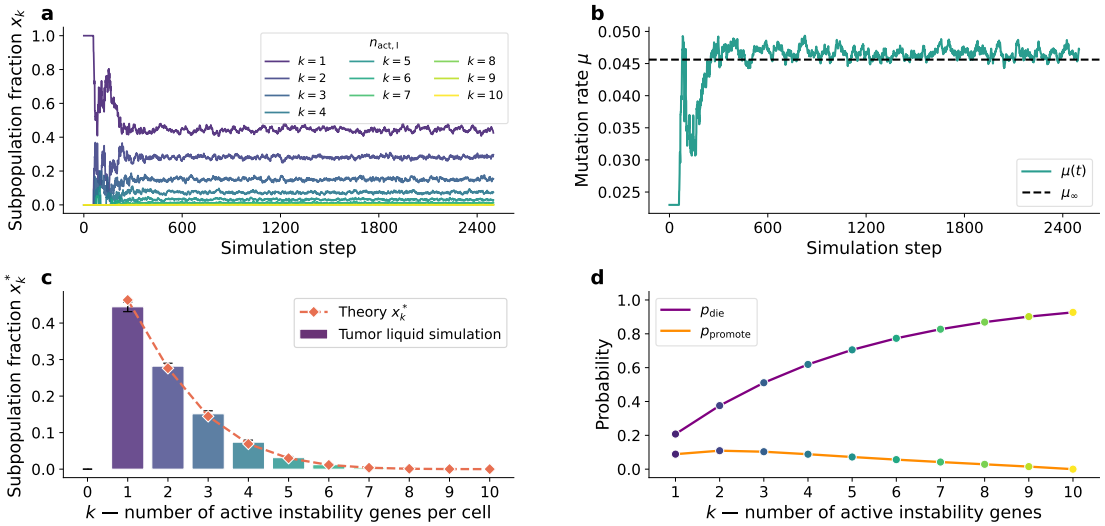


Figure 1: Evolutionary dynamics and steady-state distribution of genetic instability in liquid tumor simulations. (a) Temporal evolution of subpopulation fractions x_k as a function of simulation steps. The population is stratified by k, representing the number of active genetic instability genes per cell ($k \in [1, 10]$). Following initial transient fluctuations, the system relaxes into a stable, non-equilibrium steady state where subpopulations with low genomic instability ($k = 1, 2, 3$) dominate the tumor mass. (b) Dynamic trajectory of the global mutation rate $\mu(t)$ (solid teal line) converging toward the theoretically predicted asymptotic limit μ_∞ (dashed black line). The rapid early expansion corresponds to a transient phase of mutational acceleration before stabilizing due to selective pressures. (c) Steady-state subpopulation fractions f_k^* as a function of active instability genes. Simulated empirical distributions derived from the spatial tumor model (colored bars, error bars denote standard deviation across independent realizations) show excellent agreement with the analytical predictions derived from the mean-field framework (Eq. 41) (dashed orange line with diamonds, `Theory` x_k^*). (d) Functional dependence of phenotypic probabilities on the number of active instability genes k.

5 Mechanistic Internal-State Dynamics Across Karyotypic Regimes

To resolve the microscopic intracellular mechanisms driving the macroscopic expansion trajectories of Diploid (D, $dm = 0$), Whole-Chromosome Polyploid (P, $dm = 0.01$, whole-chromosome transfer), and Segmental Aneuploid (A, $dm = 0.01$, chunk-based transfer) tumors (Fig. 2e–f and Table I in the main manuscript), we tracked the internal cellular state variables across 500-replicate solid-tumor ensembles (Fig. 2).

5.1 Proliferation Saturation versus Effective Growth Propensity

Under the Boolean OR dominance rule governing oncogenes ($\mathcal{O}$, Eq. 239), a single mutated allele at any present copy is sufficient to activate an oncogene locus. Moreover, because initial cancer seed cells start as diploid ($N_{\mathrm{CHR}} = 2$) with zero active missegregation ($k_M = 0 \implies m = 0$), recessive inactivation of missegregation genes ($\mathcal{M}$, requiring homozygous hits on both homologues at rate $\sim \mu^2$) lags behind dominant oncogene activation (occurring at rate $\sim 2\mu$ per locus). Consequently, by the time chromosomal missegregation becomes active across the population, dominant oncogene mutations have already driven the intrinsic cellular division rate $\bar{r}(t)$ to its physiological ceiling $r_{\max} = 0.30$ across all three regimes (Fig. 2a, dashed curves). In fact, early chromosome gains slightly delay recessive tumor suppressor ($\mathcal{S}$) inactivation by increasing the number of wild-type alleles that must be mutated, and occasional chromosome losses in the mother cell can transiently strip a single-copy mutated oncogene, causing $\bar{r}(t)$ in P and A to lie marginally below D during $t \in [250, 900]$.

Thus, karyotype-dependent growth differences are not driven by raw division speed, but rather by the **effective growth propensity** (Fig. 2a, solid curves):

$$\bar{r}_{\mathrm{eff}}(t) = \bar{r}(t)\left(1 - \bar{p}_{\mathrm{d,tot}}(t)\right), \tag{42}$$

which measures the net viable reproductive output per unit time after accounting for the total per-division death probability $\bar{p}_{\mathrm{d,tot}}(t)$.

5.2 Decomposition of Per-Division Mortality: Redundancy Buffering versus Ploidy Ceiling

During cell division, mortality arises from two distinct mechanistic pathways (Fig. 2b):

1. **Essential Housekeeping ($\mathcal{H}$) Loss-of-Function ($\bar{p}_{\mathrm{d},\mathcal{H}}$, dashed curves):** A dividing lineage suffers essential gene failure if either (i) de novo point mutations simultaneously hit all remaining functional wild-type copies $k_{\mathrm{WT}}(t)$ at any of the $N_{\mathcal{H}} = 10$ essential loci, with probability

$$p_{\mathrm{d},\mathcal{H}}^{(\mathrm{mut})} = 1 - \left(1 - \mu^{k_{\mathrm{WT}}}\right)^{N_{\mathcal{H}}}, \tag{43}$$

or (ii) an asymmetric chromosomal or segmental loss (occurring with probability $m/2$ in the donor mother cell) strips the last remaining wild-type copy of a locus whose other $k - 1$ copies have already been mutated.

2. **Hyperploid ($> 5n$) and Nullisomic ($0n$) Allele-Threshold Violation ($\bar{p}_{\mathrm{d,ploidy}}$, dotted curves):** A missegregation event triggers immediate structural lethality if the recipient daughter exceeds the maximum viable copy number of 5 alleles at any locus (or if the donor loses its sole remaining chromosome copy). In Whole-Chromosome Polyploidy (P), where all loci share a uniform copy number N_{CHR}, any missegregation in a cell at $N_{\mathrm{CHR}} = 5$ pushes the daughter to $N_{\mathrm{CHR}} = 6$, yielding $\bar{p}^{(P)}_{\mathrm{d,ploidy}} = m\big(f_{n_c=5} + f_{n_c=1}\big)$. In Segmental Aneuploidy (A), where the mean effective copy number per locus plateaus at an intermediate value ($\bar{k} \approx 3.76$ across $n_c = 5$ active slots), a random chromosomal fragment only breaches the 5-copy cap if the severed segment overlaps a locus already present at 5 copies, substantially reducing the boundary hazard ($\bar{p}^{(A)}_{\mathrm{d,ploidy}} \ll \bar{p}^{(P)}_{\mathrm{d,ploidy}}$).

Combining these two contributions, $\bar{p}_{\mathrm{d,tot}} = 1 - (1 - \bar{p}_{\mathrm{d},\mathcal{H}})(1 - \bar{p}_{\mathrm{d,ploidy}})$, explains the three evolutionary regimes observed in the macroscopic tumor growth curves (Fig. 2a–b):

- **Phase I — Early Redundancy Buffering ($t \lesssim 750$):** Extra chromosome copies in P and A provide immediate allelic redundancy ($k_{\mathrm{WT}} > 2$), shielding cells from recessive $\mathcal{H}$ inactivation ($\bar{p}_{\mathrm{d},\mathcal{H}}(P) \leq \bar{p}_{\mathrm{d},\mathcal{H}}(A) < \bar{p}_{\mathrm{d},\mathcal{H}}(D)$) before a significant fraction of the population reaches the $5n$ boundary. Consequently, $\bar{r}_{\mathrm{eff}}(P) \geq \bar{r}_{\mathrm{eff}}(A) > \bar{r}_{\mathrm{eff}}(D)$.

- **Phase II — The $5n$ Ploidy Boundary Transition ($750 \lesssim t \lesssim 1450$):** Whole-chromosome gains rapidly drive the P population against the $N_{\mathrm{CHR}} = 5$ cap, causing $\bar{p}^{(P)}_{\mathrm{d,ploidy}}$ (brown dotted curve) to surge to ≈ 0.047 per division. Because Segmental Aneuploidy (A, teal dotted curve) maintains an intermediate mean copy number ($\bar{k} \approx 3.76$), its $> 5n$ boundary mortality remains more than fivefold lower (≈ 0.009), allowing A to overtake P ($\bar{r}_{\mathrm{eff}}(A) > \bar{r}_{\mathrm{eff}}(P) > \bar{r}_{\mathrm{eff}}(D)$) and establish the largest cumulative tumor mass.

- **Phase III — Stationary Buffer Saturation ($t \gtrsim 1450$):** In the long-time limit, ongoing point mutations saturate the cryptic redundancy buffer across all karyotypes ($k_{\mathrm{WT}} \to 1$; Fig. 2c). Once viable cells retain only a single unmutated copy per essential gene, point-mutation mortality converges toward the same baseline across all regimes, while ongoing missegregation imposes a permanent mortality overhead ($\bar{p}_{\mathrm{d,ploidy}}$ plus chromosomal unmasking) on P and A, yielding the asymptotic ordering $\bar{p}_{\mathrm{d,tot}}(D) < \bar{p}_{\mathrm{d,tot}}(A) < \bar{p}_{\mathrm{d,tot}}(P)$ and $\bar{r}_{\mathrm{eff}}(D) > \bar{r}_{\mathrm{eff}}(A) > \bar{r}_{\mathrm{eff}}(P)$.

5.3 Cryptic Recessive Damage Accumulation and Buffer Erosion

Figure 2c directly quantifies the internal genetic load buffered by multi-copy redundancy via the fraction of present essential housekeeping alleles that are mutated ($\mathrm{mut}_{\mathcal{H}}$) in surviving cells. Because complete inactivation of any essential locus is lethal, a viable cell with k copies at an essential locus can harbor at most $k - 1$ mutated alleles. In Diploid (D) cells ($k = 2$), $\mathrm{mut}_{\mathcal{H}}$ is strictly bounded by the viability ceiling $(2 - 1)/2 = 0.50$ (black dotted line). By contrast, Whole-Chromosome Polyploid (P) and Segmental

Aneuploid (A) cells survive while accumulating a massive cryptic reservoir of inactivated essential alleles, saturating at $(5-1)/5 = 0.80$ in P and ≈ 0.76 in A. Once this $(k-1)/k$ redundancy buffer is completely filled, the protective survival advantage of polyploidy and aneuploidy is exhausted, converting chromosomal instability from an early adaptive buffer into a late-stage liability.

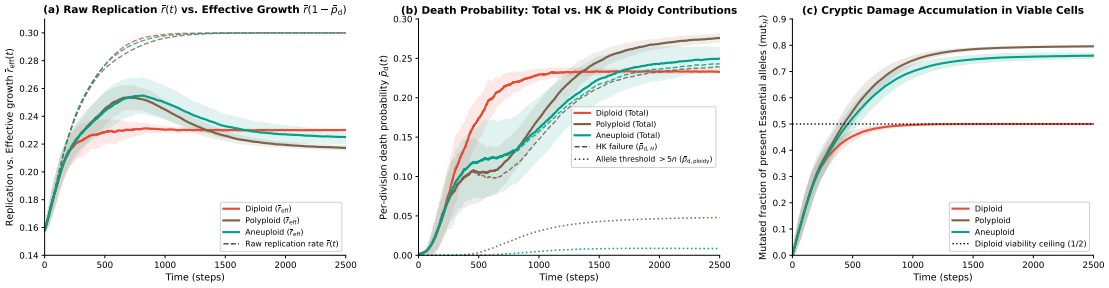


Figure 2: **Internal-state mechanistic dynamics across karyotypic regimes in solid tumors. (a)** Raw cellular replication rate $\bar{r}(t)$ (dashed curves) versus effective growth propensity $\bar{r}_{\text{eff}}(t) = \bar{r}(t)(1-\bar{p}_{\text{d,tot}}(t))$ (solid curves). Because $\bar{r}(t)$ rapidly saturates at $r_{\max} = 0.30$ across all regimes, inter-karyotype growth differences are governed by differential cellular survival rather than intrinsic division speed. **(b)** Decomposition of the per-division death probability $\bar{p}_{\text{d,tot}}(t)$ (solid curves) into essential housekeeping ($\mathcal{H}$) loss-of-function lethality ($\bar{p}_{\text{d},\mathcal{H}}$, dashed curves) and hyperploid ($> 5n$) allele-threshold lethality ($\bar{p}_{\text{d,ploidy}}$, dotted curves). **(c)** Mutated fraction of present essential housekeeping alleles ($\text{mut}_{\mathcal{H}}$) in surviving cells, demonstrating cryptic recessive damage accumulation above the diploid viability ceiling ($1/2 = 0.50$, black dotted line) up to the polyploid ($4/5 = 0.80$) and aneuploid (≈ 0.76) saturation limits. Curves and shaded bands denote ensemble means ± 1 s.d. across surviving tumor realizations (500 replicates per regime).